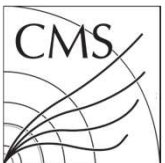


Reconstructing $H \rightarrow \tau\tau$ – events with AK15 jets

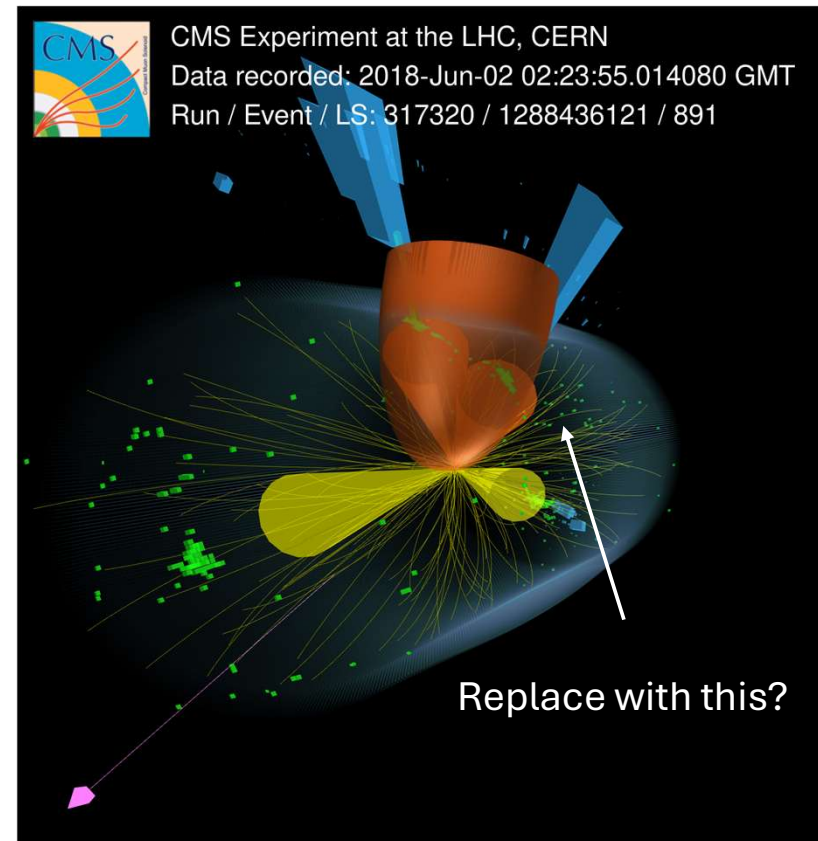
Milko Roine

SHiP 26.6.2026



Goal

- In standard topology (< 250 GeV) $H \rightarrow \tau\tau$ process reconstructed with 2 AK4 jets
- In highly boosted topologies (> 450 GeV) use one AK8 jet
- Could AK15 jets provide improved performance in the semi-boosted area of 300 – 500 GeV?



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Method

1) Compare matching efficiencies in MC simulated data

- Gluon fusion to pure $H \rightarrow \tau\tau$ (pynthia 8)
- Vector boson fusion to $HH \rightarrow b\bar{b}\tau\tau$ (pynthia 8)



2) Compare tagging efficiencies in MC simulated data by combining ParticleNet, UParT and DeepTau taggers with a BDT



3) Proof of Concept -LHC data-analysis

Jet Preselection

- $|\eta| < 2.5$
- In detector frame τ separation $\sim 2m_H/p_T$
 - Missing p_T from neutrinos suggests should cut lower
- Based on distribution of good jets:

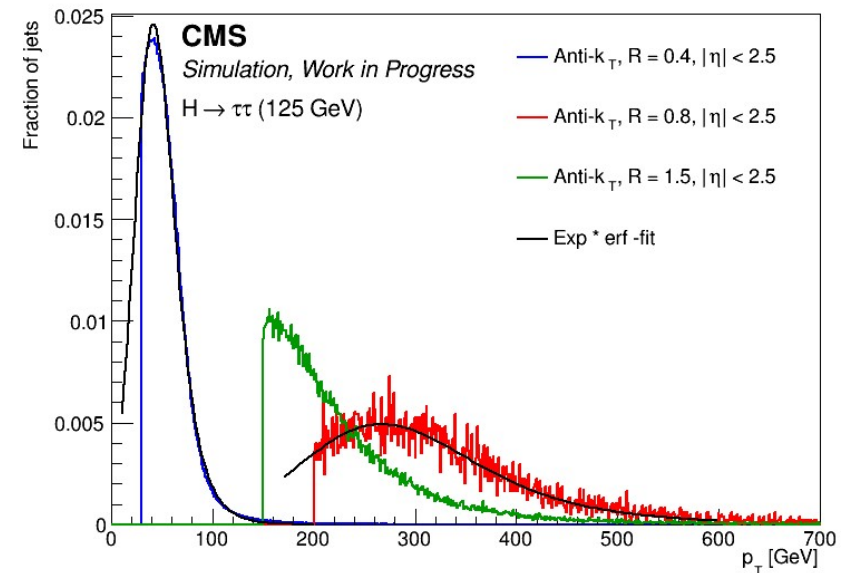
AK4: $p_T \geq 30$ GeV, $|\eta| < 2.5$

AK8: $p_T \geq 200$ GeV, $|\eta| < 2.5$

AK15: $p_T \geq 150$ GeV, $|\eta| < 2.5$

$$p_T \gtrsim \frac{2m_H}{\Delta R}$$

- AK8: $p_T \geq 312$ GeV
- AK15: $p_T \geq 167$ GeV



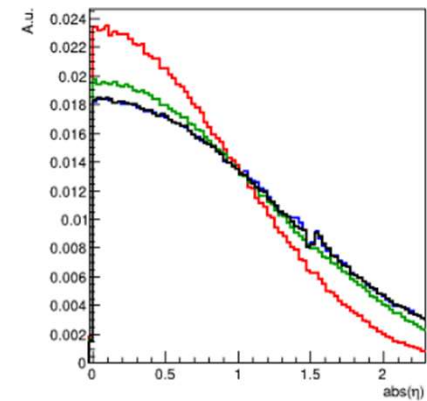
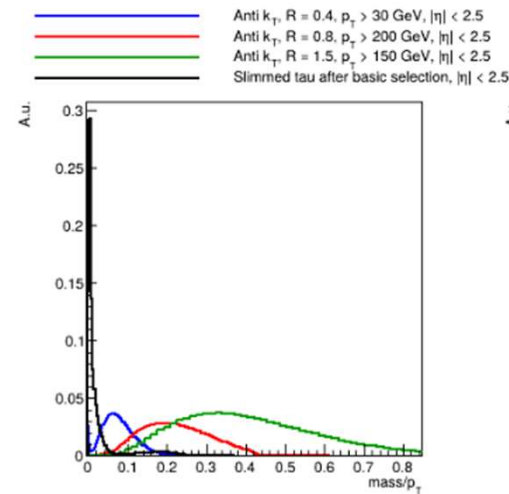
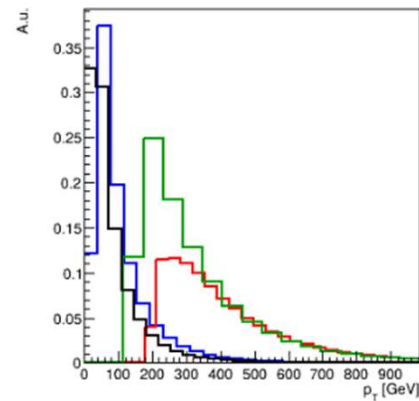
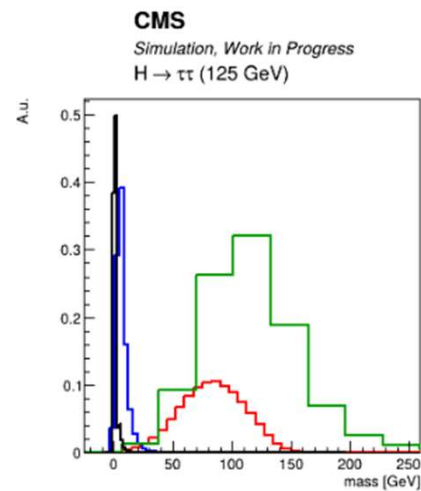
Skimming

1. Truth labels

- Find hard-process last-copy τ from Higgs
- Require hadronic decay

2. Jet Matching

- Require opposite charge taus with the same hard-process Higgs ancestor inside ΔR cone
- Only keep events with exactly one good jet



Results

AK15 has superior matching efficiency everywhere > 300 GeV

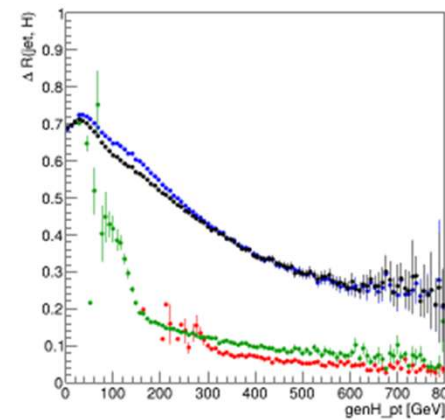
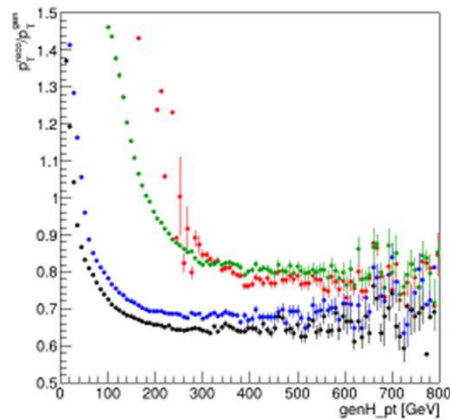
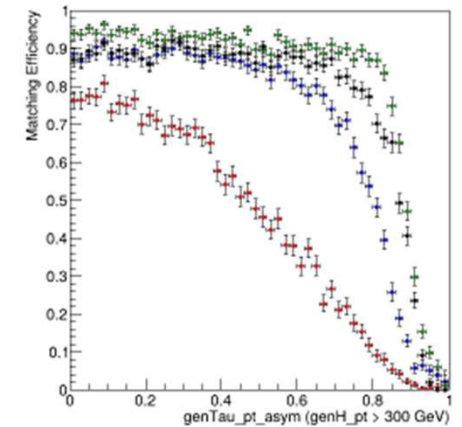
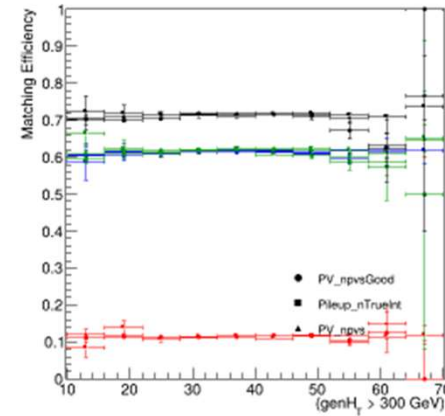
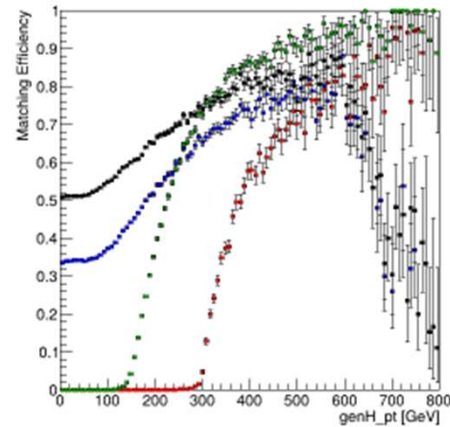
Higher tolerance of tau asymmetry

$$\frac{|p_{T,\tau1} - p_{T,\tau2}|}{p_{T,\tau1} + p_{T,\tau2}}$$

in > 300 GeV topologies

Energy response
 $p_T^{reco.}/p_T^{Gen.}$ converges slightly faster than fatJet

13.6 TeV, powheg-pythia8,
 GluGluH → 2Tau



CMS

Simulation, Work in Progress

H → ττ (125 GeV)

- Anti k_T, R = 0.4, p_T > 30 GeV, |η| < 2.5
- Anti k_T, R = 0.8, p_T > 200 GeV, |η| < 2.5
- Anti k_T, R = 1.5, p_T > 150 GeV, |η| < 2.5
- Skimmed taus after basic selection, |η| < 2.5

Results

AK15 has superior matching efficiency everywhere > 300 GeV

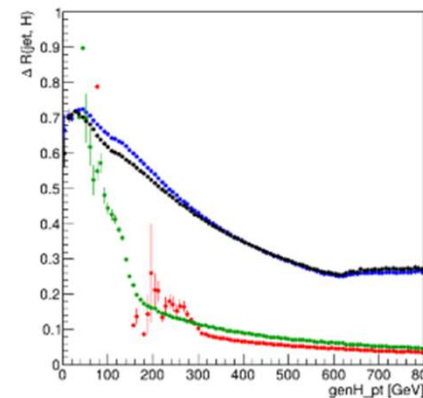
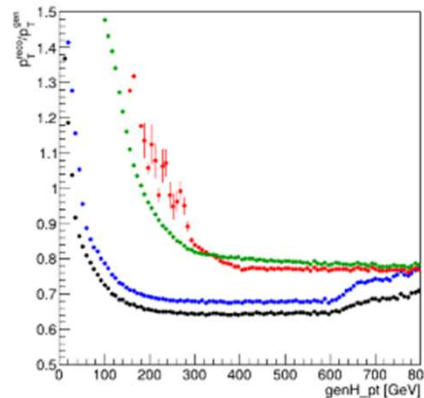
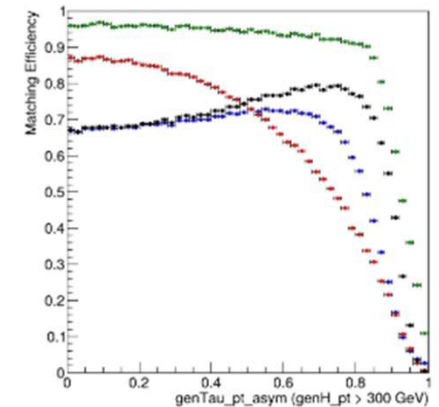
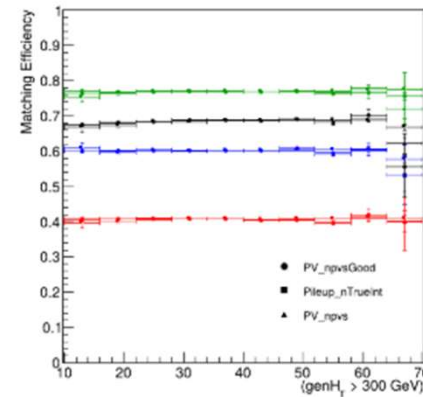
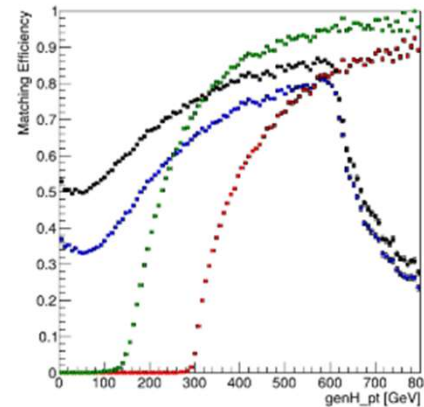
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$$\frac{|p_{T,\tau 1} - p_{T,\tau 2}|}{p_{T,\tau 1} + p_{T,\tau 2}}$$

in > 300 GeV topologies

Energy response
 $p_T^{reco.}/p_T^{Gen.}$ converges slightly faster than fatJet

13.6 TeV, madgraph-pythia8,
 VBFHH → 2B2Tau



CMS

Simulation, Work in Progress

H → ττ (125 GeV)

- Anti k_T, R = 0.4, p_T > 30 GeV, |η| < 2.5
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- Anti k_T, R = 1.5, p_T > 150 GeV, |η| < 2.5
- Skimmed taus after basic selection, |η| < 2.5

Analysis of subjects

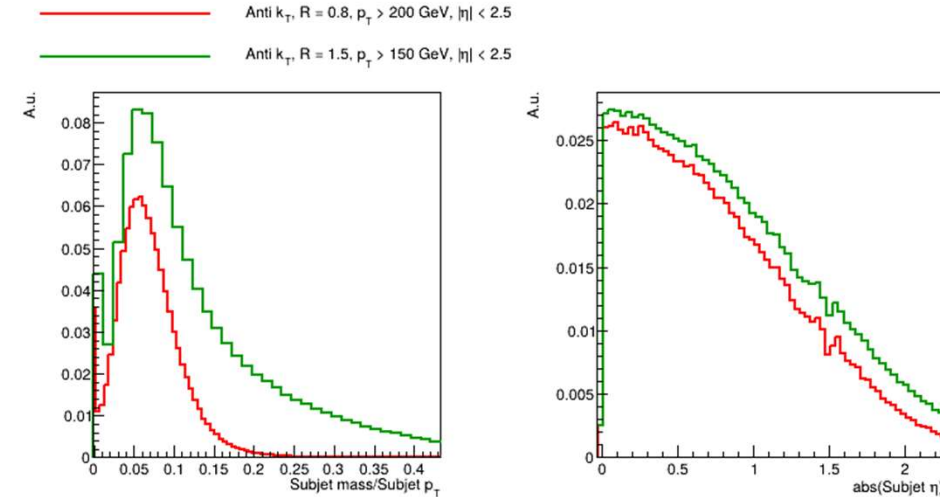
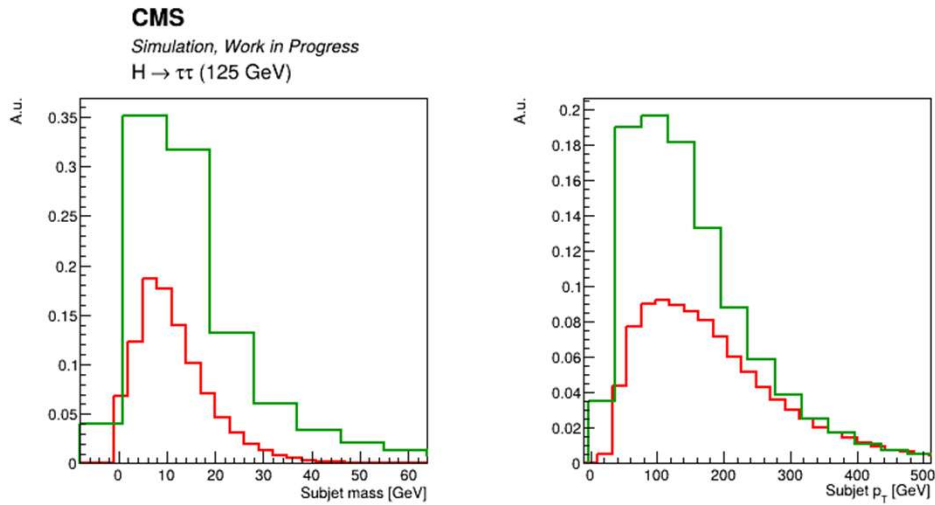
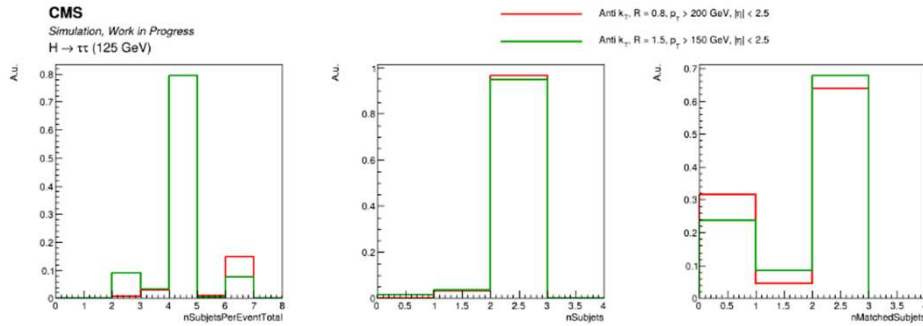
Subjet matching

1. Subjet preselection

- Require jets inside a matched parent jet ($\Delta R(jet, parent) < \Delta R_{jet}$)

2. Jet Matching

- Require opposite charge taus with the same hard-process Higgs ancestor inside ΔR cone
- Tight selection: Require both subjects to be matched to genTruth tau pair
- Loose selection: Require at least one subjet matched

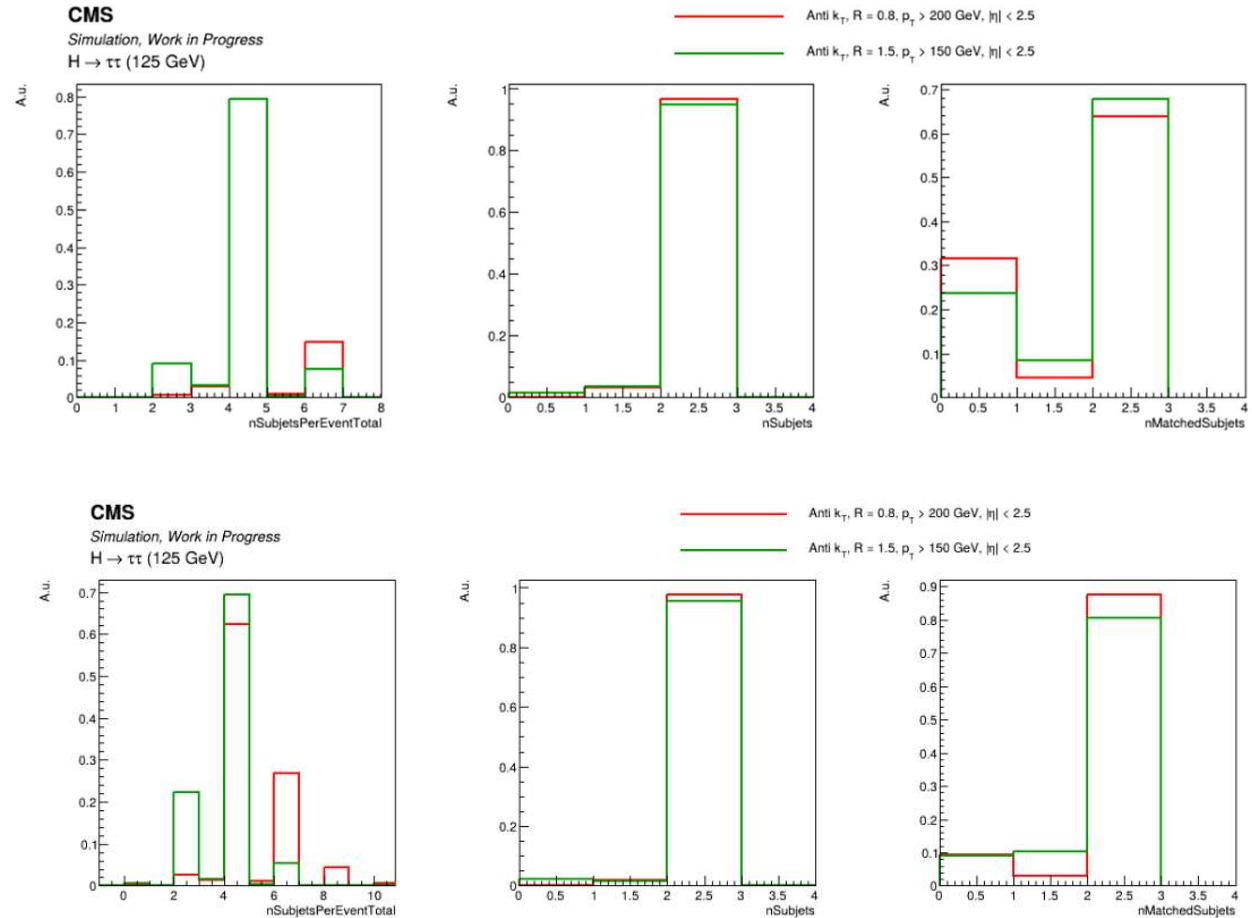


Note

13.6 TeV, madgraph-pythia8,
VBFHH $\rightarrow$ 2B2Tau

Difference between
simulation methods?

13.6 TeV, powheg-pythia8,
GluGluH $\rightarrow$ 2Tau



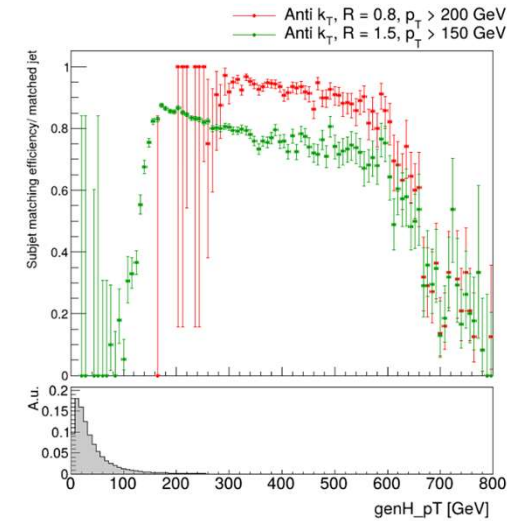
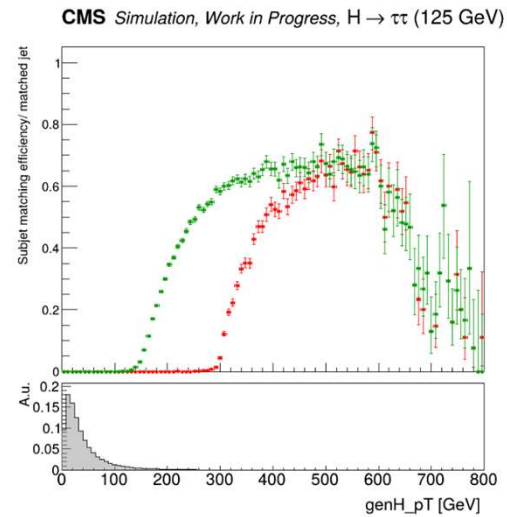
Matching efficiency

Against all events: AK15
superior everywhere

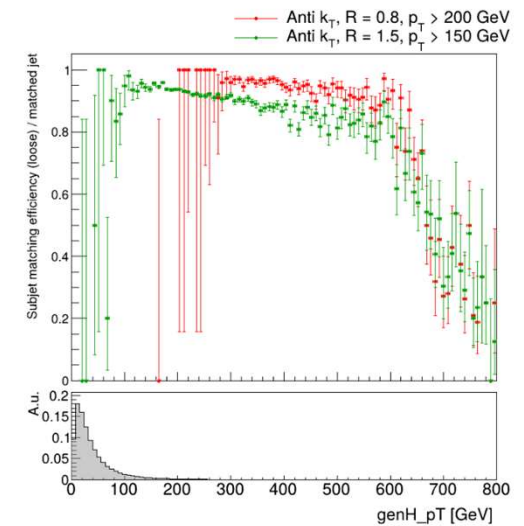
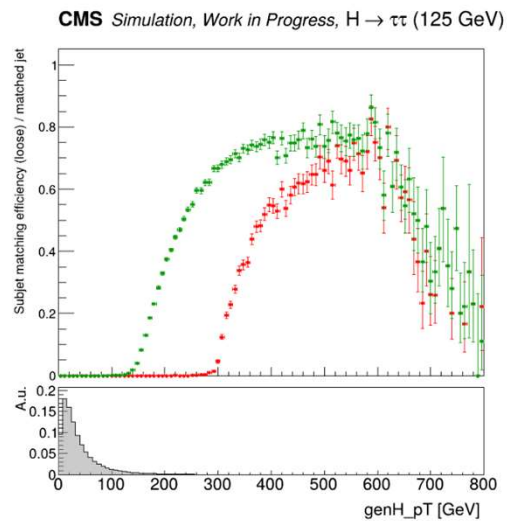
Against already matched
parent jets: AK8 slightly
more performant

13.6 TeV, powheg-pythia8,
GluGluH $\rightarrow$ 2Tau

Tight



Loose



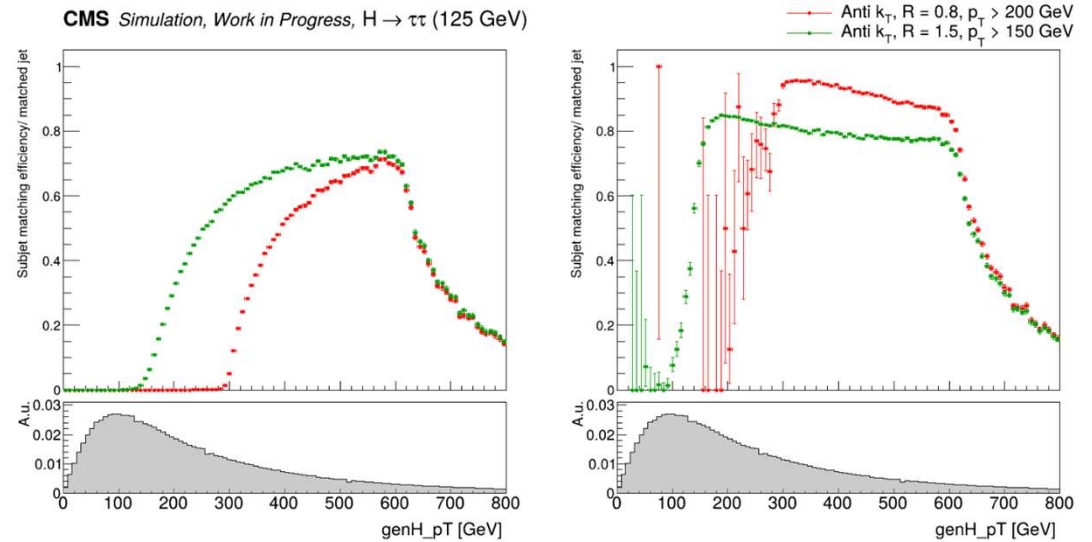
Matching efficiency

Against all events: AK15
superior everywhere

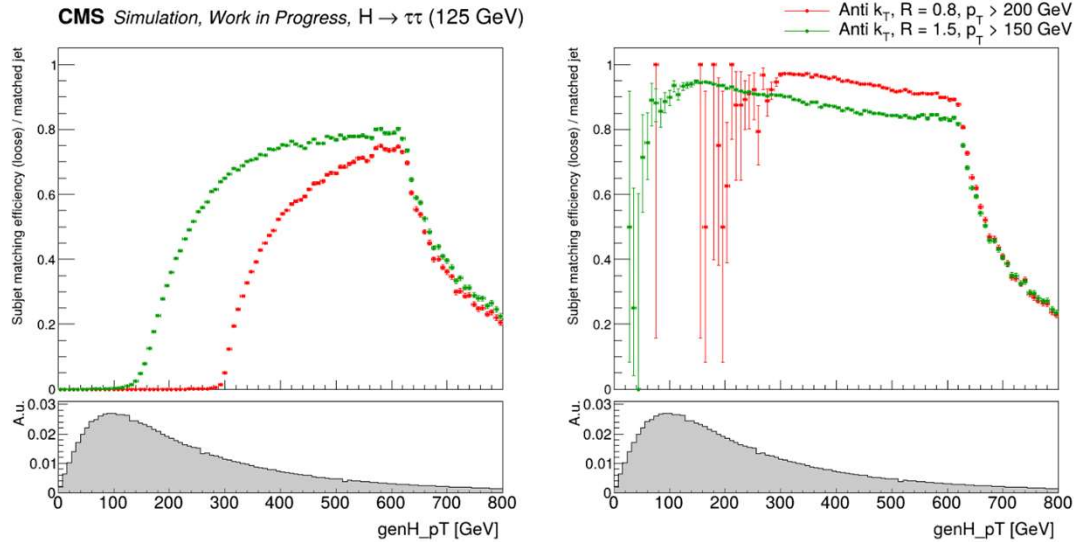
Against already matched
parent jets: AK8 slightly
more performant

13.6 TeV, madgraph-pythia8,
VBFHH $\rightarrow$ 2B2Tau

Tight



Loose



Side quest: Jet clustering kinematics at

$$dR(\tau_1, \tau_2) \approx \Delta R_{jet}$$

Minmax -problem

- To study matching efficiency in more detail, define observable

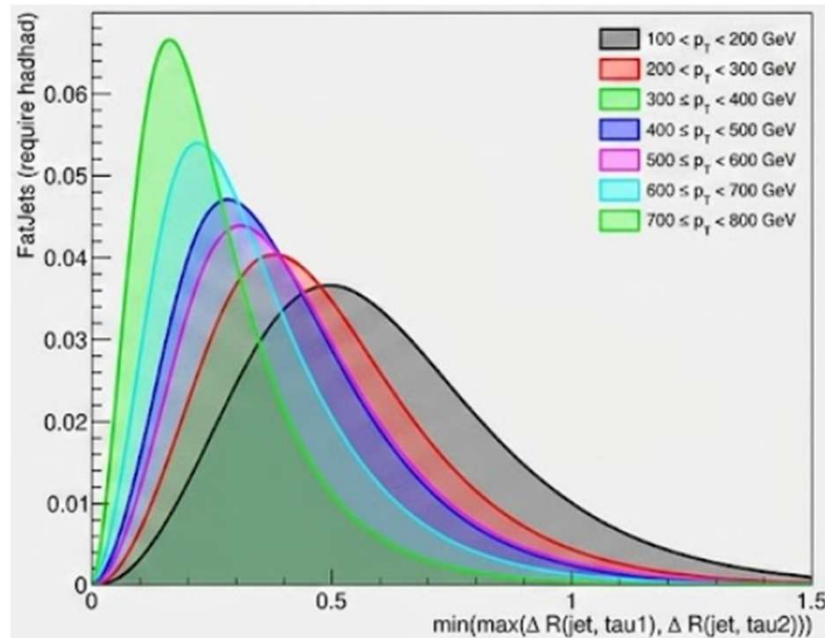
$$\min_{jets} \max_{\tau_1, \tau_2} \{dR(\tau_1, jet), dR(\tau_2, jet)\}$$

where τ_1, τ_2 are the genTruth taus.

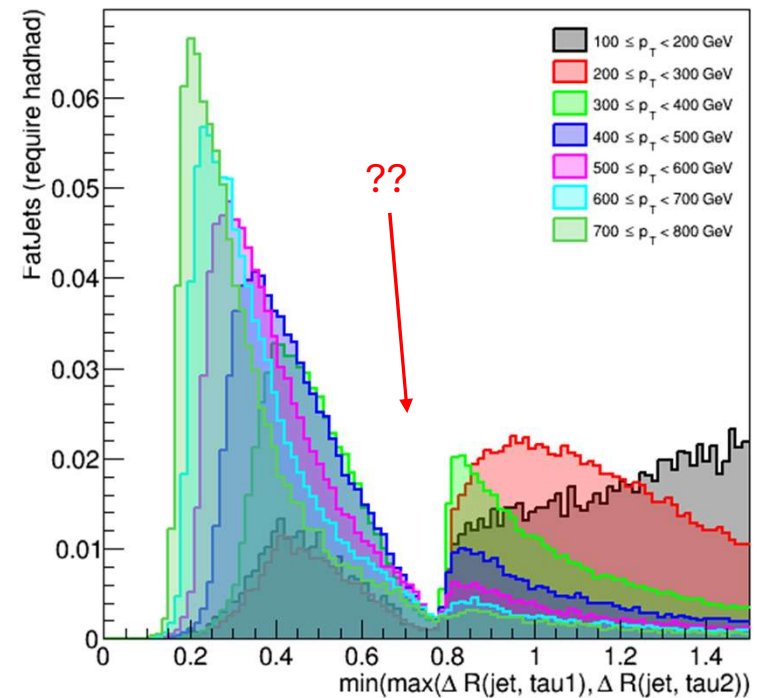
- I.e. choose the best jet and calculate its distance to the furthest tau
 - Idea: Visualize the distribution of good and bad jets at different p_T -scales

Minmax -problem

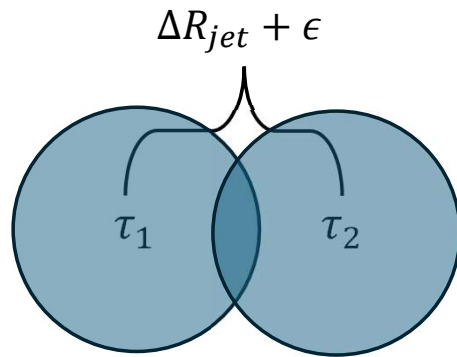
Expected:



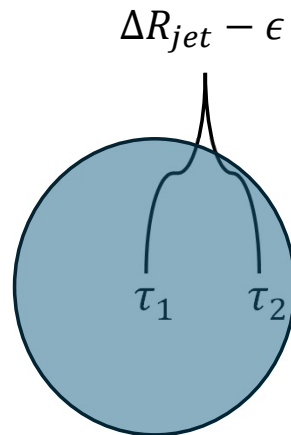
Got:



Explanation?

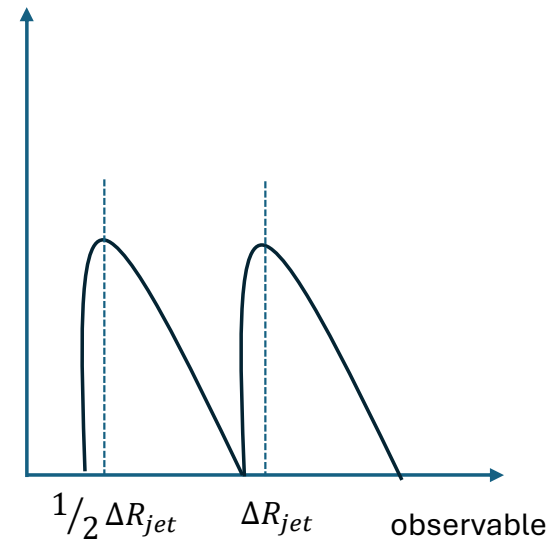
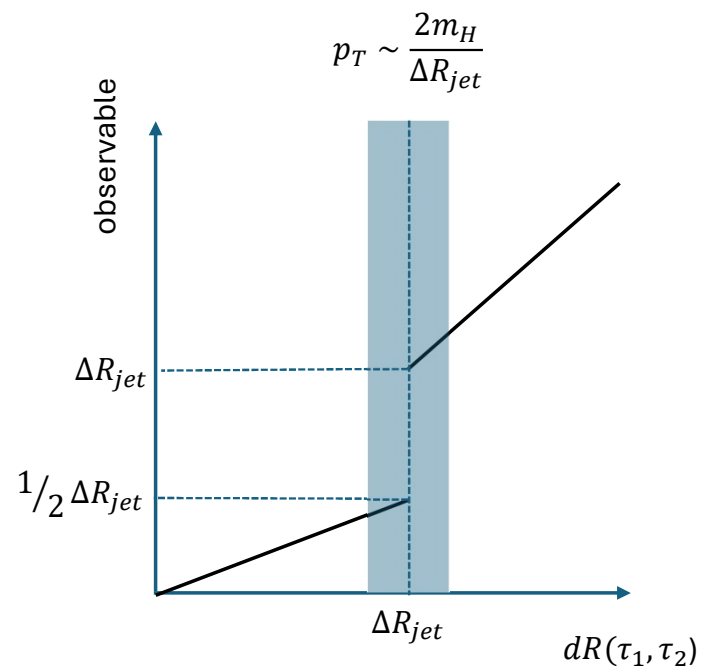


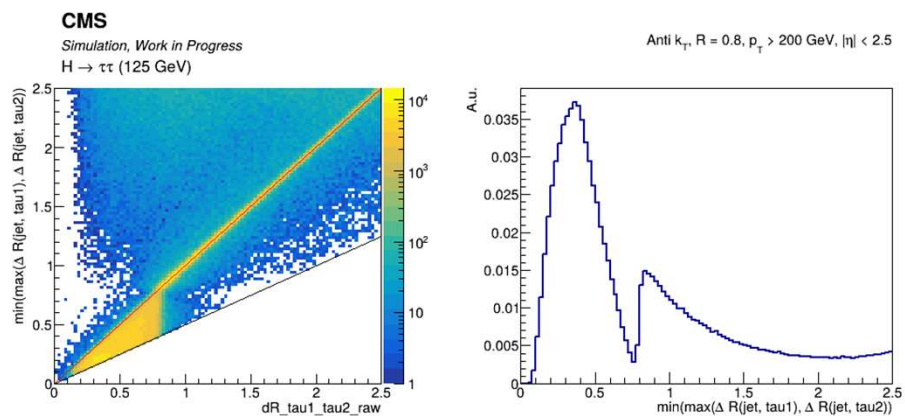
$$\min_{jets} \max_{\tau_1, \tau_2} \{dR(\tau_1, jet), dR(\tau_2, jet)\} \approx dR(\tau_1, \tau_2)$$



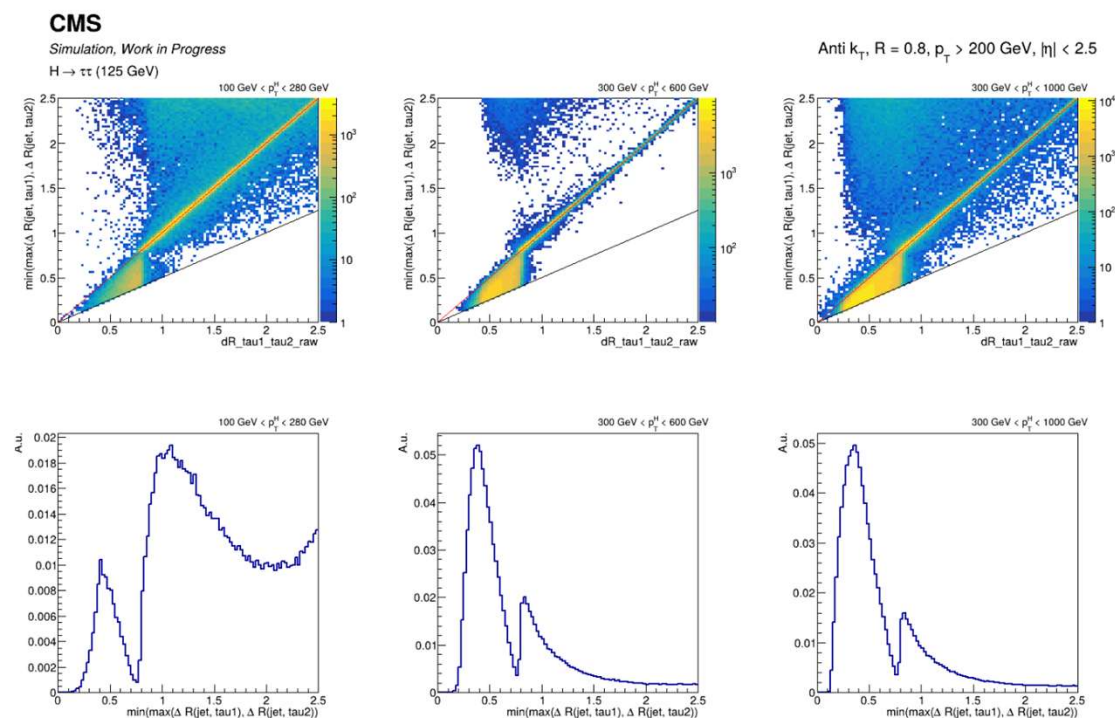
$$\min_{jets} \max_{\tau_1, \tau_2} \{dR(\tau_1, jet), dR(\tau_2, jet)\} \approx \frac{1}{2} dR(\tau_1, \tau_2)$$

Explanation?



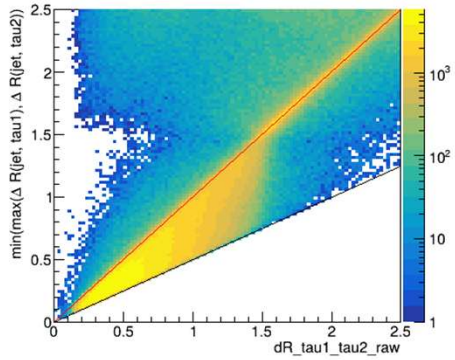


Explanation?

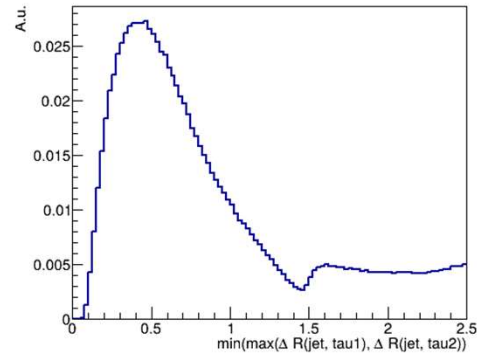


CMS

Simulation, Work in Progress
 $H \rightarrow \tau\tau$ (125 GeV)



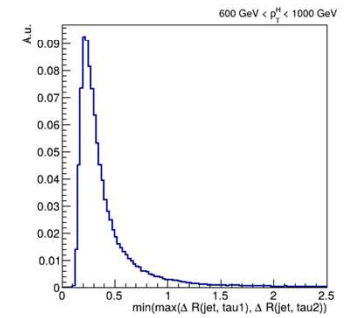
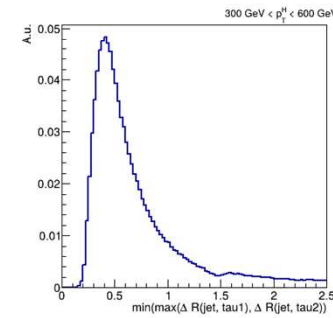
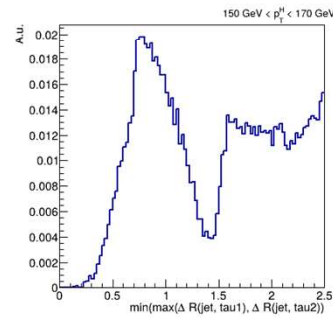
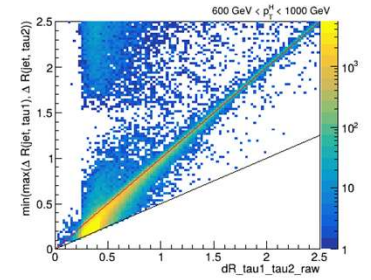
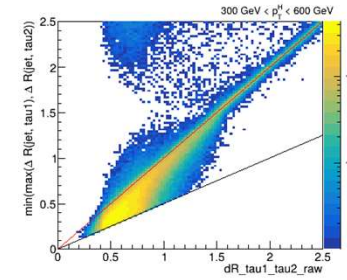
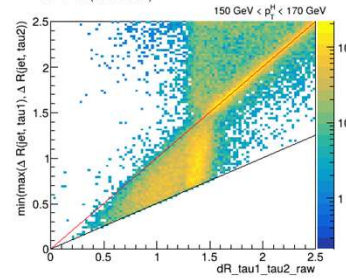
Anti k_T , $R = 1.5$, $p_T > 150$ GeV, $|\eta| < 2.5$



Explanation?

CMS

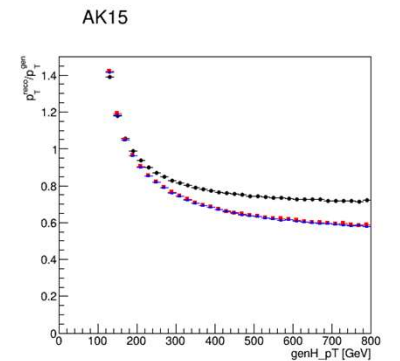
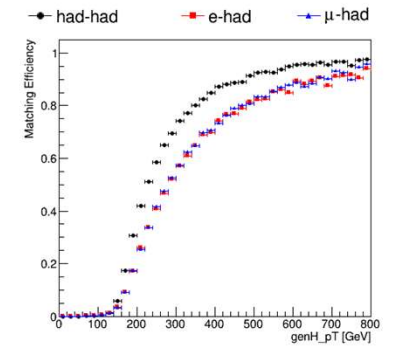
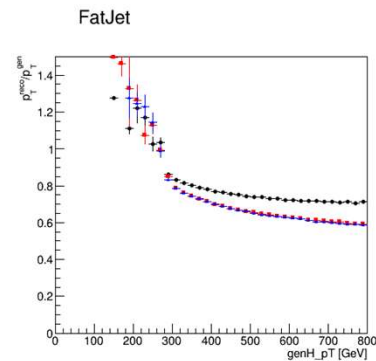
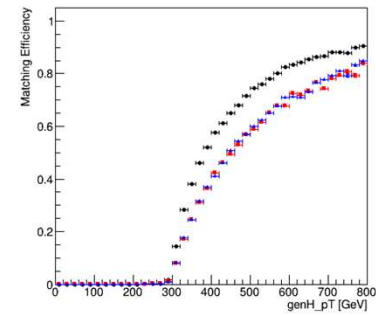
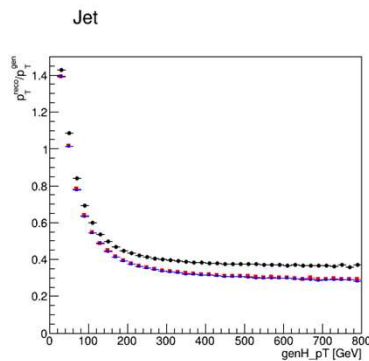
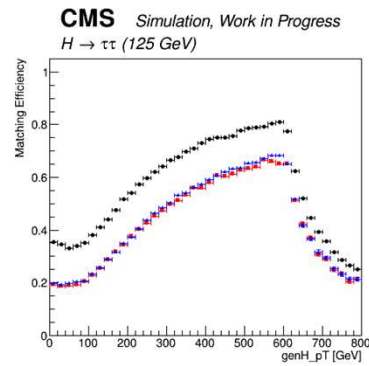
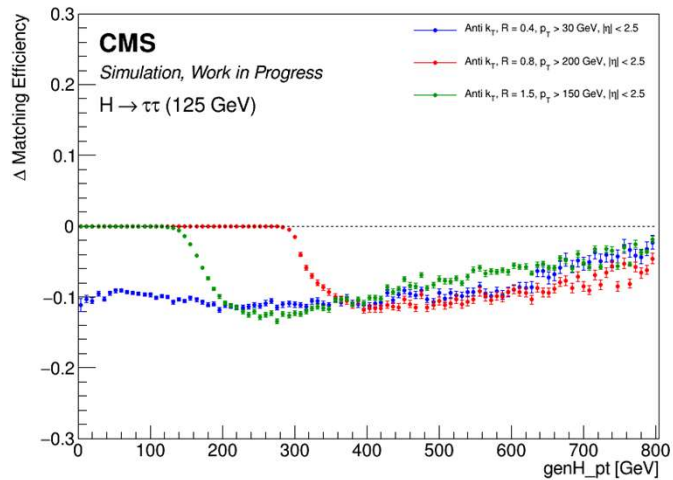
Simulation, Work in Progress
 $H \rightarrow \tau\tau$ (125 GeV)



26.6.2026

Side quest 2: Effect of decay channel on matching efficiency

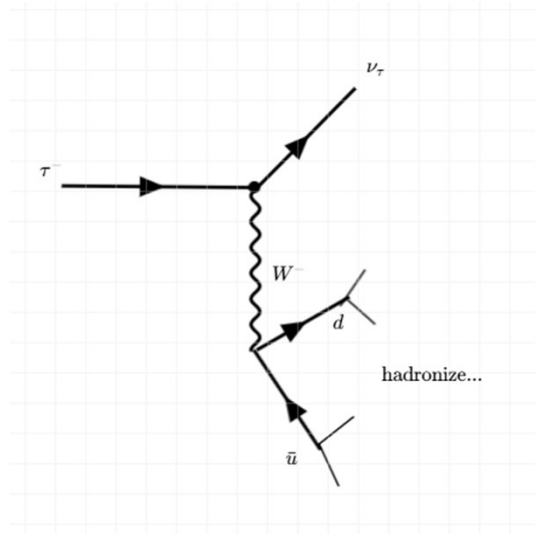
Semileptonic decay channels
cause $\sim 10\%$ decay in jet
performance. **Why?**



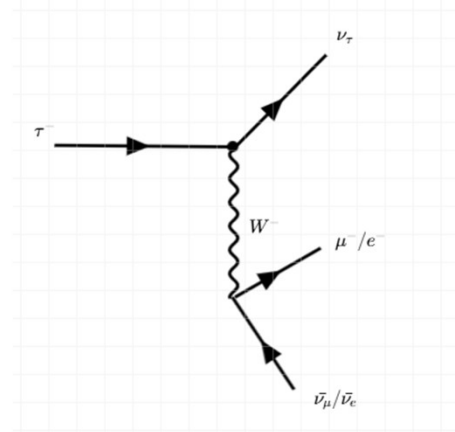
Energy accounting

Branching ratios:
Hadronic $\sim 65\%$
Semileptonic $\sim 35\%$

[[PDG Review 58, "τ Branching Fractions"](#)]



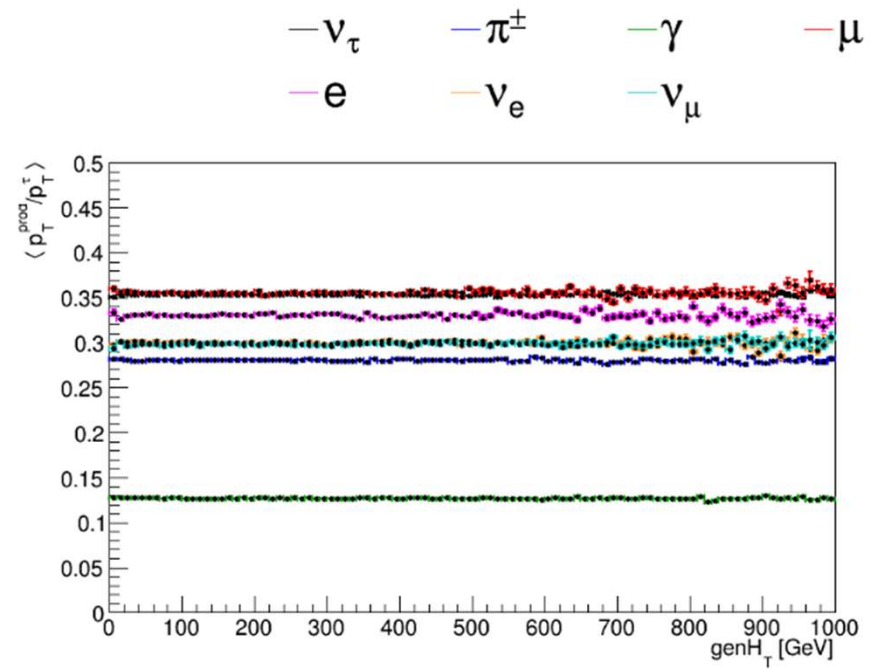
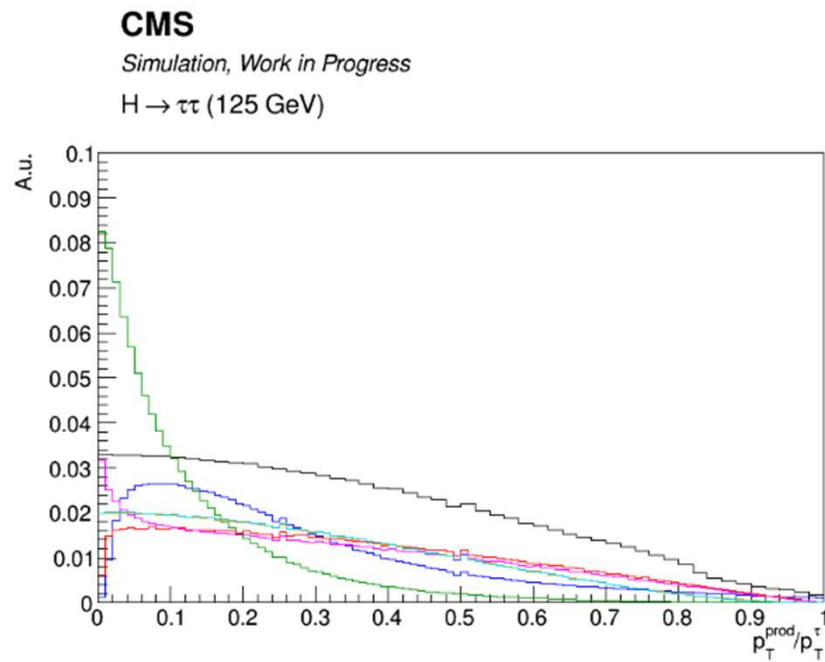
$\sim 1/3$ of the energy
invisible



$\sim 2/3$ of the energy
invisible

Does the energy actually split evenly?

- Yes, but I don't know why (yet)



Energy accounting

Hadronic:

$$\left\langle \frac{p_T^{jet}}{p_T^{reco.}} \right\rangle \approx \frac{2}{3} + \frac{C}{p_T^H}$$

$$C \propto \rho A_{jet}$$

Semileptonic:

$$\left\langle \frac{p_T^{jet}}{p_T^{reco.}} \right\rangle \approx \frac{1}{2} \left(\frac{2}{3} + \frac{1}{3} \right) + \frac{C}{p_T^H} = \frac{1}{2} + \frac{C}{p_T^H}$$

Branching ratios:
Hadronic $\sim 65\%$
Semileptonic $\sim 35\%$

[[PDG Review 58, "τ Branching Fractions"](#)]

All:

$$\left\langle \frac{p_T^{jet}}{p_T^{reco.}} \right\rangle \approx 2 \cdot 0.35 \cdot 0.65 \cdot \frac{1}{2} + 0.65^2 \cdot \frac{2}{3} + \frac{C}{p_T^H} = 0.509 + \frac{C}{p_T^H}$$

Energy accounting

Fit

$$\frac{p_T^{reco}}{p_T^{gen}} = f + \frac{C}{genH_pt}$$

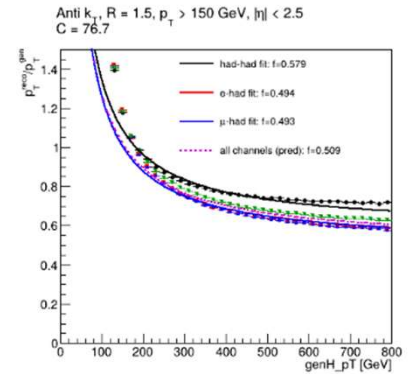
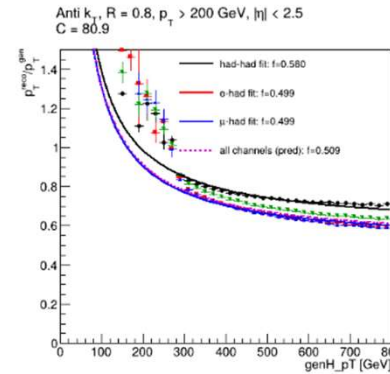
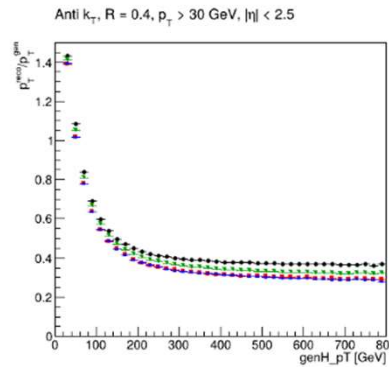
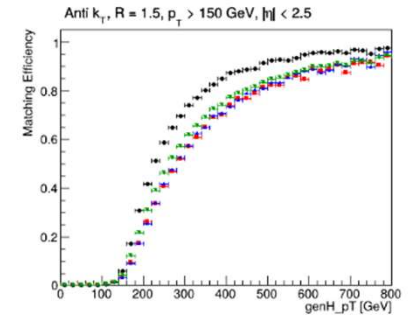
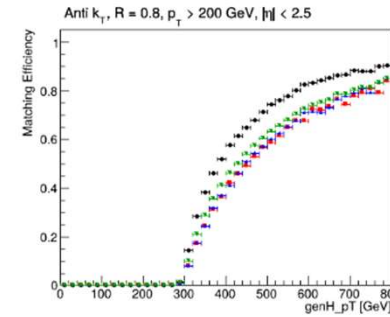
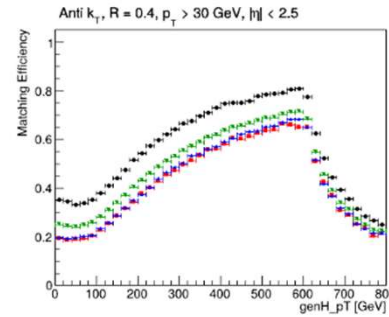
with f free per channel
and C free per jet type

Channel	Predicted	Fit
Hadronic	$f = 0.66$	$f = 0.580 \pm 0.001$
μ -had	$f = 0.5$	$f = 0.499 \pm 0.005$
e^- -had	$f = 0.5$	$f = 0.499 \pm 0.006$

Sanity check: Plot prediction with fitted C
and $f = 0.509$ against data with all channels

CMS Simulation, Work in Progress
 $H \rightarrow \tau\tau$ (125 GeV)

• had-had • e-had • μ -had • all channels



Matching efficiency

Turn on curve: Must be



Predict change in matching efficiency when including semileptonic events:

$$p_T^{jet} \approx f_{channel} p_T^H \geq p_T^{cut}$$

$$\rightarrow p_T^H \geq (f_{channel})^{-1} p_T^{cut}$$

$$\Delta\epsilon(p_T^H) \approx 0.65^2 \left(\epsilon_{hadhad} \left(\frac{f_{all}}{f_{hadhad}} p_T^H \right) - \epsilon_{hadhad}(p_T^H) \right)$$

$$\approx 0.43 (\epsilon_{hadhad}(0.77 p_T^H) - \epsilon_{hadhad}(p_T^H))$$

Hadronic:

$$p_T^H \geq \frac{p_T^{cut}}{0.66} \approx 1.52 p_T^{cut}$$

Semileptonic:

$$p_T^H \geq \frac{p_T^{cut}}{0.5} \approx 2 p_T^{cut}$$

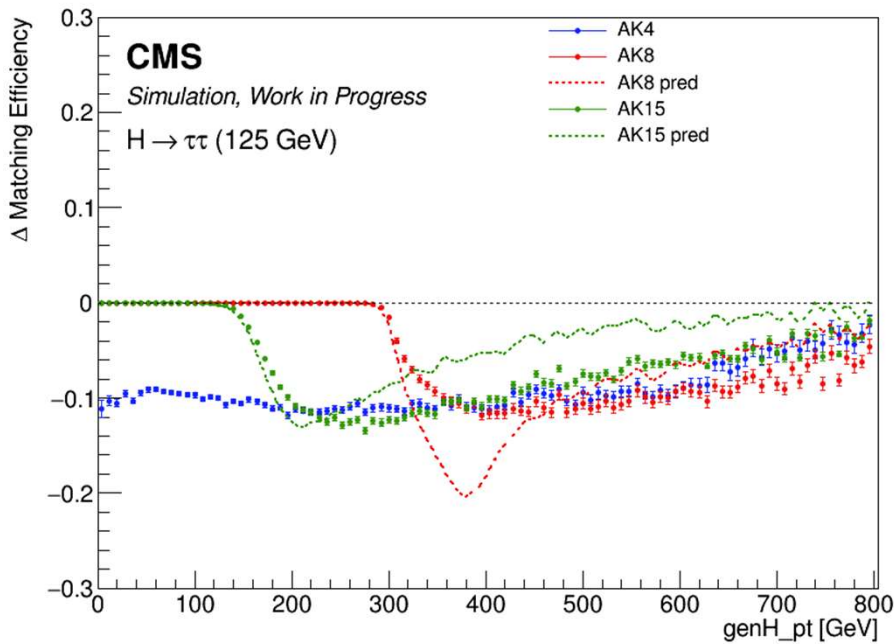
Branching ratios:

Hadronic ~ 65%

Semileptonic ~ 35%

[PDG Review 58, "[τ Branching Fractions](#)"]

Matching efficiency



Predict change in matching efficiency when including semileptonic events:

$$\Delta\epsilon(p_T^H) \approx 0.65^2 \left(\epsilon_{hadhad} \left(\frac{f_{all}}{f_{hadhad}} p_T^H \right) - \epsilon_{hadhad}(p_T^H) \right)$$

$$\approx 0.43(\epsilon_{hadhad}(0.77p_T^H) - \epsilon_{hadhad}(p_T^H))$$

Predicts turn-on, but geometric effects that persist in high pt not captured

Matching efficiency: correction

$$\epsilon \approx P(\Delta R(\tau_1, \tau_2) \leq \Delta R_{jet}) P(p_T^{jet} \geq p_T^{cut})$$



Only accounted for this



$$\approx \sqrt{1 - 4 \left(\frac{m_H}{p_T^H \Delta R_{jet}} \right)^2}$$

Better model of tau separation:

$$\Delta R(\tau_1, \tau_2) \approx \frac{m_H}{p_T^H} (z(1-z))^{-1/2} \leq \Delta R_{jet}$$

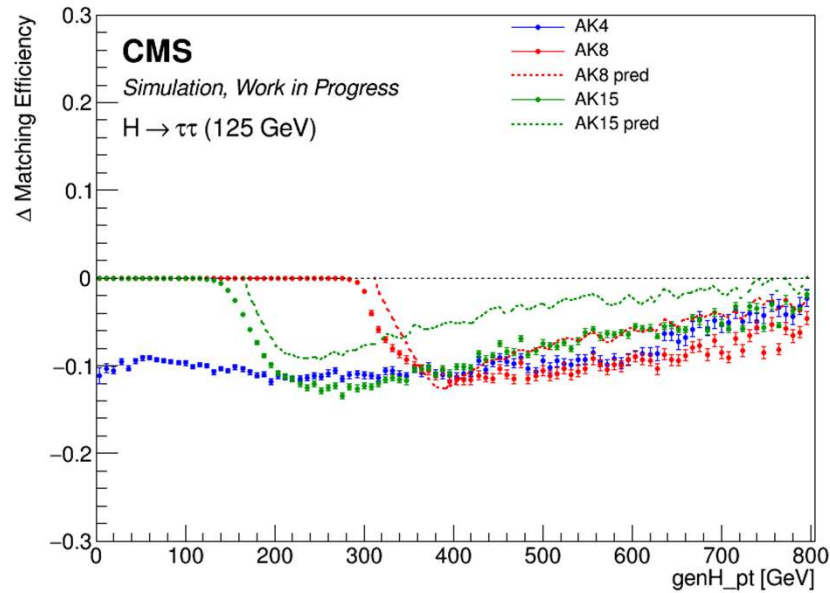
where $z = \frac{p_T^{\tau_1}}{p_T^H}$. Solving for z gives

$$-1/2 \sqrt{1 - 4 \left(\frac{m_H}{p_T^H \Delta R_{jet}} \right)^2} \leq z \leq 1/2 \sqrt{1 - 4 \left(\frac{m_H}{p_T^H \Delta R_{jet}} \right)^2}$$

Assuming z is uniform the propability of this is

$$\sqrt{1 - 4 \left(\frac{m_H}{p_T^H \Delta R_{jet}} \right)^2}$$

Matching efficiency: correction



Predict change in matching efficiency when including semileptonic events:

$$\Delta\epsilon(p_T^H) \approx 0.43 \sqrt{1 - 4 \left(\frac{m_H}{p_T^H \Delta R_{jet}} \right)^2} (\epsilon_{hadhad}(0.77 p_T^H) - \epsilon_{hadhad}(p_T^H))$$

Next...

- Analysis of Tagging efficiency in progress
- Expansion of subjet structure analysis